Topological Data Analysis

Lecture 7

Higher-order Laplacian

Oleg Kachan

Graph Laplacian

Graph

$G = (V, E)$, where $E \subseteq V \times V$.

Adjacency matrix

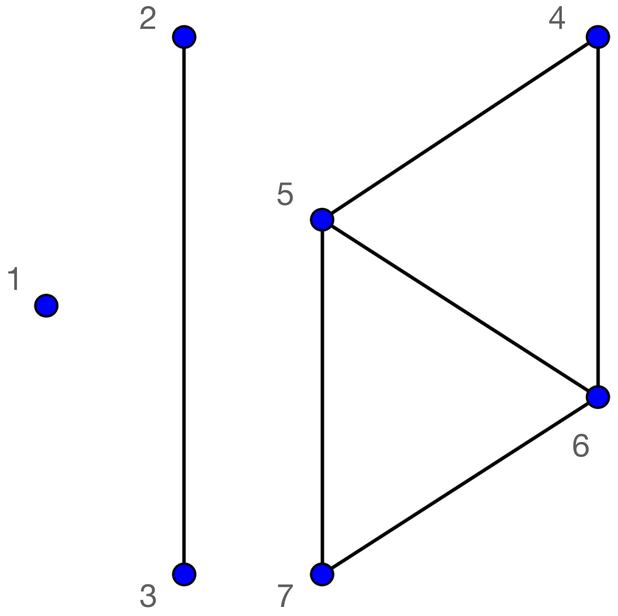
A is |V| x |V| matrix.

$$\mathbf{A}_{ij} = \begin{cases} 1, & v_i \sim v_j, \\ 0, & \text{otherwise}. \end{cases}$$

Incidence matrix

B is |V| x |E| matrix.

$$\mathbf{B}_{ij} = \begin{cases} 1, & v_i \sim v_j, v_i \succ v_j, \\ -1, & v_i \sim v_j, v_i \prec v_j, \\ 0, & \text{otherwise}. \end{cases}$$



Graph *G*

Graph Laplacian

Graph

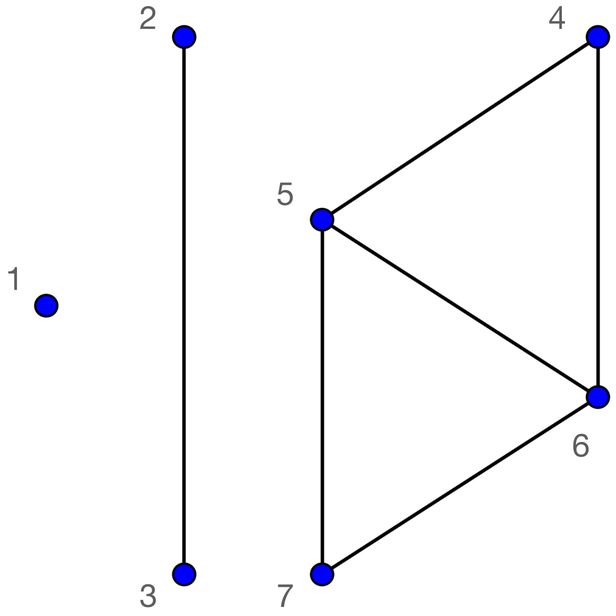
$G = (V, E)$, where $E \subseteq V \times V$.

Laplacian matrix

L is |V| x |V| matrix.

$$\mathbf{L}_{ij} = \begin{cases} d(v_i), & i = j, \\ -1, & i \neq j \text{ and } v_i \sim v_j, \\ 0, & \text{otherwise}. \end{cases}$$

	1	2	3	4	5	6	7
1	0						
2		1	-1				
3		-1	1				
4				2	-1	-1	
5				-1	3	-1	-1
6				-1	-1	3	-1
7					-1	-1	2



Graph *G*

Laplacian via adjacency

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

Laplacian via incidence

$$\mathbf{L} = \mathbf{B}\mathbf{B}^T$$

Graph Laplacian

$$(BB^T)_{ij} = \sum_{k=1}^{n} B_{ik}B_{jk}$$

$$B_{ik}B_{jk} = \begin{cases} 1, & i = j, (i,j) \in E, \\ -1, & i \neq j, (j,i) \in E, \\ 0, & \text{overwise}\,. \end{cases} \qquad (BB^T)_{ij} = \begin{cases} \sum_{k|i\in E_k} 1, & i = j, \\ -1, & i \neq j, \\ 0, & \text{overwise}\,. \end{cases}$$

$$\sum_{k|i\in E_k} 1 = \deg(v_i)$$ *is the degree of i-th vertex, therefore* $\mathbf{BB}^T = \mathbf{D} - \mathbf{A}$

Graph Laplacian

Properties

- real symmetric
- rows/columns sums to 0
- positive-semidefinite, all eigenvalues >=0, eigenvalues are real

- $\lambda_0 = 0$, as $\boldsymbol{v}_0 = (1, 1, \ldots, 1)^T$ satisfies $\boldsymbol{L}\boldsymbol{v}_0 = 0$
- the number of connected components of G is the dimension of the nullspace (kernel) of L

Higher-order Laplacian

Simplicial complex

$$K = \left(\Sigma_0, \Sigma_1, \ldots, \Sigma_{\dim(K)}\right)$$

Chain space

$$C_k = \left\{a\sigma_k \mid a \in \mathbb{F}, \sigma_k \in \Sigma_k\right\}$$

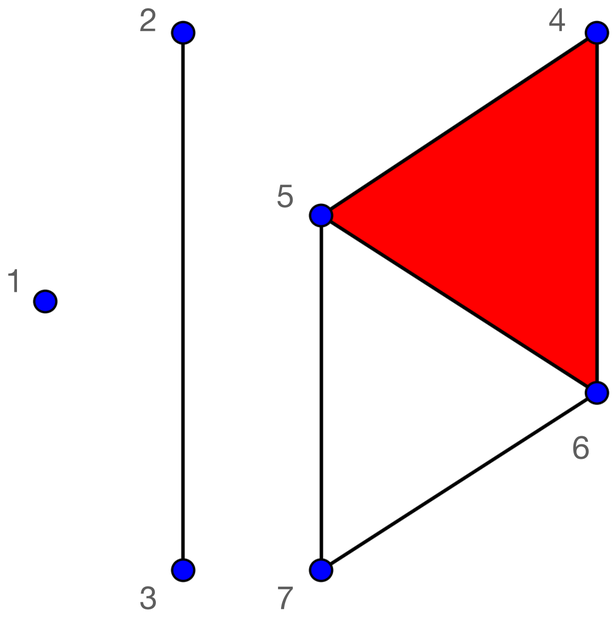
Boundary operator

$$\partial_k : C_k \to C_{k-1}$$

$$\partial_k([v_0, \ldots, v_k]) = \sum_{i=0}^{k} (-1)^i [v_0, \ldots, \hat{v}_i, \ldots, v_k]$$

$$\partial_1([2,3]) = 3 - 2$$

$$\partial_2([4,5,6]) = [5,6] - [4,6] + [4,5]$$



Simplicial complex *K*

$\mathbf{B}_1 =$

	23	45	46	56	57	67
1						
2	-1					
3	1					
4		-1	-1			
5		1		-1	-1	
6			1	1		-1
7					1	1

$\mathbf{B}_2 =$

	456
23	
45	1
46	-1
56	1
57	
67	

Higher-order Laplacian

Chain complex of *K*

$$\dots \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

$$\partial_k \circ \partial_{k+1} = 0$$

Higher-order Laplacian operator

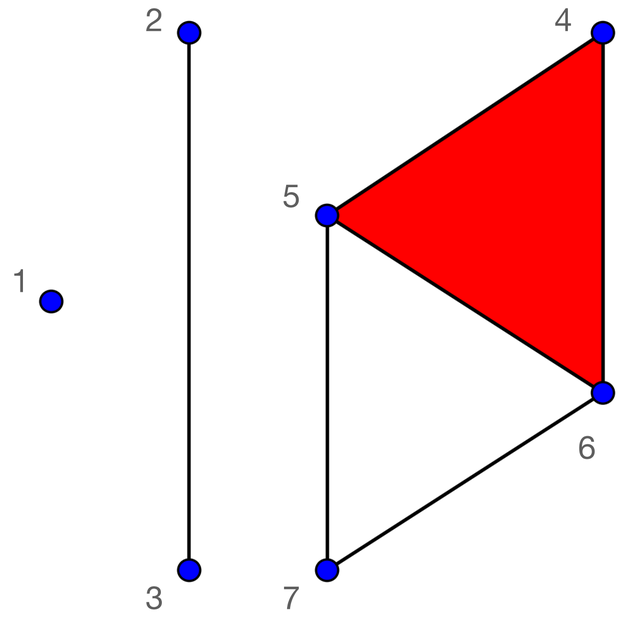
$$L_k = \partial_k^* \partial_k + \partial_{k+1} \partial_{k+1}^*$$

Spectra of Combinatorial Laplace Operators on Simplicial Complexes
Horak D., Jost J. *Chaos, Advances in Mathematics (2013)*

Higher-order Laplacian

The multiplicity of zero eigenvalue of $\boldsymbol{L}_k$ equals the rank of the k-th homology group H_k of K.

$$\dim \ker(\mathbf{L}_k) = H_k(K)$$

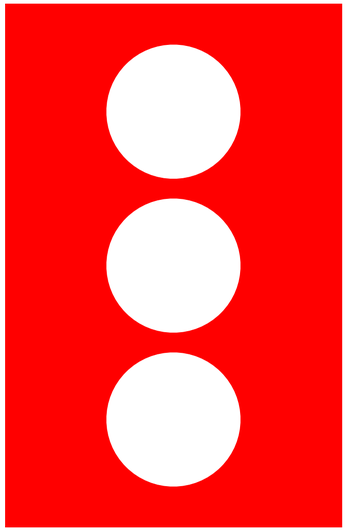


Simplicial complex *K*

Spectra of Combinatorial Laplace Operators on Simplicial Complexes
Horak D., Jost J. *Chaos, Advances in Mathematics (2013)*

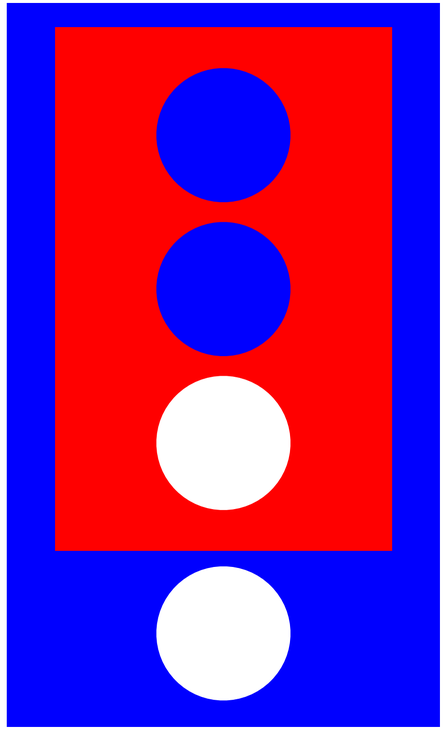
Persistent Laplacian

K



$\hookrightarrow$

$K \subset L$



$\beta_1(K) = 3$

$\beta_1(L) = 2$

$\beta_1(K \subset L) = 1$

Persistent Laplacian

$\iota : K \hookrightarrow L$

$$\begin{tikzcd}
C_{p+1}^{K} \arrow[r, "\partial_{p+1}^{K}"] \arrow[d, hook, dashed] & C_{p}^{K} \arrow[d, hook, dashed] \arrow[r, shift right, "\partial_{p}^{K}"'] & C_{p-1}^{K} \arrow[l, shift right, "(\partial_{p}^{K})^{*}"'] \arrow[d, hook, dashed] \\
C_{p+1}^{L} \arrow[r, "\partial_{p+1}^{L}"] & C_{p}^{L} \arrow[r, "\partial_{p}^{L}"] \arrow[l, shift right, "\partial_{p}^{L,K}"'] & C_{p-1}^{L} \\
C_{p+1}^{L,K} \arrow[u, hook, dashed] \arrow[ur, shift right, "(\partial_{p+1}^{L,K})^{*}"'] & &
\end{tikzcd}$$

$$C_{q+1}^{K\subset L} = \{c \in C_{q+1}^{L} \mid \partial_{q+1}^{L}(c) \in C_{q}^{K}\} \subseteq C_{q+1}^{L} \qquad \partial_{q+1}^{K\subset L} := \partial_{q+1}^{L}|_{C_{q+1}^{K\subset L}} : C_{q+1}^{K\subset L} \to C_{q}^{K}$$

$$\mathbf{L}_k(K \subset L) = \mathbf{L}_k^{UP}(K \subset L) + \mathbf{L}_k^{DOWN}(K) = \mathbf{L}_k^{UP}(L) \;/\; \mathbf{L}_k^{UP}(L \backslash K) + \mathbf{L}_k^{DOWN}(K)$$

Persistent Laplacians: Properties, Algorithms and Implications
Memoli F. et al. *(2022)*

Persistent Laplacian

$$\iota : K \hookrightarrow L$$

$$C_{q+1}^{L,K} = \{c \in C_{q+1}^{L} \mid \partial_{q+1}^{L}(c) \in C_{q}^{K}\}$$

$$\partial_{q+1}^{L,K} := \partial_{q+1}^{L}|_{C_{q+1}^{L,K}} : C_{q+1}^{L}, K \to C_{q}^{K}$$

$$\Delta_{q,UP}^{L,K} := \partial_{q+1}^{L,K} \circ \left(\partial_{q+1}^{L,K}\right)^{*}$$

$$\Delta_{q,DOWN}^{K} := \left(\partial_{q}^{K}\right)^{*} \circ \partial_{q}^{K}$$

$$\Delta_{q}^{L,K} := \Delta_{q,UP}^{L,K} + \Delta_{q,DOWN}^{K}$$

Persistent Laplacians: Properties, Algorithms and Implications
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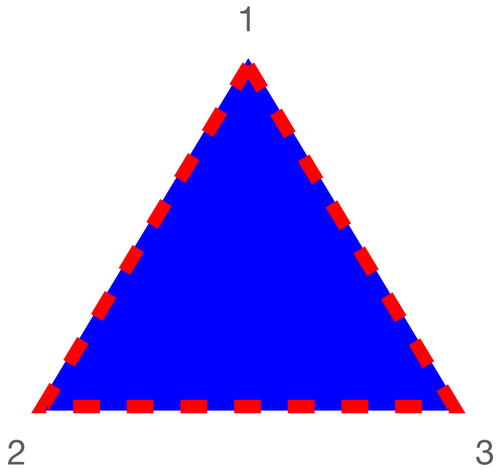
Persistent Laplacian

Triangle



$K \subset L$

$\beta_1(K) = 1$

$\beta_1(L) = 0$

$\beta_1(K \subset L) = 0$

$$\mathbf{L}_1(K \subset L) = \mathbf{L}_1^{UP}(K \subset L) + \mathbf{L}_1^{DOWN}(K)$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) \ / \ \mathbf{L}_k^{UP}(L \backslash K)$$

$$\mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T$$

$$\mathbf{L}_1^{DOWN}(K) = \mathbf{B}_1(K)^T\mathbf{B}_1(K)$$

$\mathbf{B}_1(K) =$ $\mathbf{B}_1(L) =$

	12	13	23
1	-1	-1	
2	1		-1
3		1	1

$\mathbf{B}_2(L) =$

	123
12	-1
13	1
23	

$\mathbf{L}_1^{DOWN}(K) =$

	12	13	23
1	2	1	-1
2	1	2	1
3	-1	1	2

$\mathbf{L}_1^{UP}(K \subset L) =$

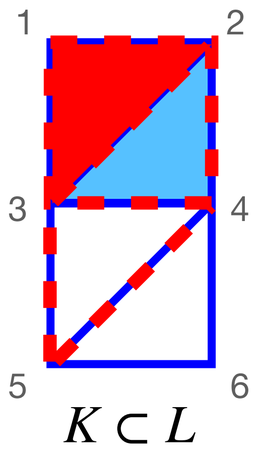
	12	13	23
1	1	-1	1
2	-1	1	-1
3	1	-1	1

$\mathbf{L}_1(K \subset L) =$

	12	13	23
1	3		
2		3	
3			3

Persistent Laplacian

Truss



$\beta_1(K) = 2$

$\beta_1(L) = 2$

$\beta_1(K \subset L) = 1$

$$\mathbf{L}_1(K \subset L) = \mathbf{L}_1^{UP}(K \subset L) + \mathbf{L}_1^{DOWN}(K)$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) \ / \ \mathbf{L}_k^{UP}(L \backslash K)$$

$$\mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T$$

$$\mathbf{L}_1^{DOWN}(K) = \mathbf{B}_1(K)^T\mathbf{B}_1(K)$$

$\mathbf{B}_1(K)$

	12	13	23	24	34	35	45
1	-1	-1					
2	1		-1	-1			
3		1	1		-1	-1	
4				1	1		-1
5						1	1

$\mathbf{L}_1^{DOWN}(K)$

	1	2	3	4	5
1					
2					
3					
4					
5					

Persistent Laplacian

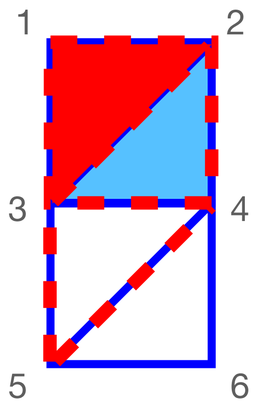
Truss

$$\mathbf{L}_1(K \subset L) = \mathbf{L}_1^{UP}(K \subset L) + \mathbf{L}_1^{DOWN}(K)$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) \ / \ \mathbf{L}_k^{UP}(L \backslash K)$$

$$\mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T \qquad \mathbf{L}_1^{DOWN}(K) = \mathbf{B}_1(K)^T\mathbf{B}_1(K)$$



$K \subset L$

$\beta_1(K) = 2$

$\beta_1(L) = 2$

$\beta_1(K \subset L) = 1$

$\mathbf{B}_2(L) =$

	012	123
01	1	
02	-1	
12	1	1
13		-1
23		1
24		
34		
35		
45		

$\mathbf{L}_1^{UP}(L) =$

	01	02	12	13	23	24	34	35	45
01									
02									
12									
13									
23									
24									
34									
35									
45									

Schur complement

Given a block matrix

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

The Schur complement of D in M is defined

$$M/D = A - BD^{\dagger}C$$

Where $D^{\dagger}$ is the pseudoinverse of D.

Analogously, the Schur complement of A in M

$$M/A = D - BA^{\dagger}C$$

Persistent Laplacians: Properties, Algorithms and Implications
Memoli F. et al. *(2022)*

Persistent Laplacian

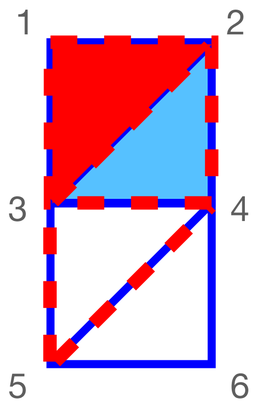
Truss

$$\mathbf{L}_1(K \subset L) = \mathbf{L}_1^{UP}(K \subset L) + \mathbf{L}_1^{DOWN}(K)$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) \ / \ \mathbf{L}_k^{UP}(L \backslash K)$$

$$\mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T$$

$$\mathbf{L}_1^{UP}(K \subset L) = \mathbf{L}_k^{UP}(L) = \mathbf{B}_2(L)\mathbf{B}_2(L)^T \qquad \mathbf{L}_1^{DOWN}(K) = \mathbf{B}_1(K)^T\mathbf{B}_1(K)$$



$K \subset L$

$\beta_1(K) = 2$

$\beta_1(L) = 2$

$\beta_1(K \subset L) = 1$

$\mathbf{B}_2(L) =$

	012	123
01	1	
02	-1	
12	1	1
13		-1
23		1
24		
34		
35		
45		

$\mathbf{L}_1^{UP}(L) =$

	01	02	12	13	23	24	34	35	45
01									
02									
12									
13				A				B	
23									
24									
34									
35			$C = B^T$					$D = L\backslash K$	
45									

$\mathbf{L}_1^{UP}(K \subset L) =$
$A - BD^{-1}C$

	01	02	12	13	23	24	34
01							
02							
12							
13							
23							
24							
34							

Generalized higher-order Laplacian

Generalized chain complex of *K*

$$\begin{tikzcd} C_3 \arrow[r, "\partial_{3,1}"] \arrow[rr, bend right, "\partial_{3,3}"'] & C_2 \arrow[r, "\partial_{2,1}"] \arrow[r, bend right, "\partial_{3,2}"'] & C_1 \arrow[r, "\partial_{1,1}"] \arrow[r, bend right, "\partial_{2,2}"'] & C_0 \end{tikzcd}$$

Generalized Laplacian operator

$$L_{k,p,q} = \partial^*_{k,p}\partial_{k,p} + \partial_{k+q,q}\partial^*_{k+q,q}$$

Simplicial Degree in Complex Networks. Applications of Topological Data Analysis to Network Science
Serrano D.H. et al. *Chaos, Solitons & Fractals (2020)*

Generalized higher-order Laplacian

Generalized chain complex of *K*

$$\begin{tikzcd} C_3 \arrow[r, "\partial_{3,1}"] \arrow[rr, bend right, "\partial_{3,2}"'] \arrow[rrr, bend right=60, "\partial_{3,3}"'] & C_2 \arrow[r, "\partial_{2,1}"] \arrow[rr, bend right, "\partial_{2,2}"'] & C_1 \arrow[r, "\partial_{1,1}"] & C_0 \end{tikzcd}$$

Generalized boundary operator

$$\partial_{k,p} : C_k \to C_{k-p}$$

$$\partial_{k,p}([v_{\eta(0)}, \ldots, v_{\eta(k)}]) = \sum_{j_1,\ldots,j_p} \mathrm{sgn}(\eta)\mathrm{sgn}(\varepsilon_{j_1\ldots j_p})[v_0, \ldots, \hat{v}_{j_1}, \ldots, \hat{v}_{j_p}, \ldots, v_k]$$

Given the (k-p)-face τ of k-simplex σ = $[v_0, \ldots, \hat{v}_{j_1}, \ldots, \hat{v}_{j_p}, \ldots, v_k]$ *denote the permutation*

$$\varepsilon_{j_1\ldots j_p} = \begin{pmatrix} 0 & \ldots & p-1 & k & \ldots & k \\ j_1 & \ldots & j_h & 1 & \ldots & 1 \end{pmatrix}$$

Simplicial Degree in Complex Networks. Applications of Topological Data Analysis to Network Science
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Higher-order Laplacian

Spectrum

Higher-order Laplacian operator

$$C_2 \underset{\partial_2^*}{\overset{\partial_2}{\rightleftarrows}} C_1 \underset{\partial_1^*}{\overset{\partial_1}{\rightleftarrows}} C_0$$

$$L_k = \partial_k^* \partial_k + \partial_k \partial_k^*$$

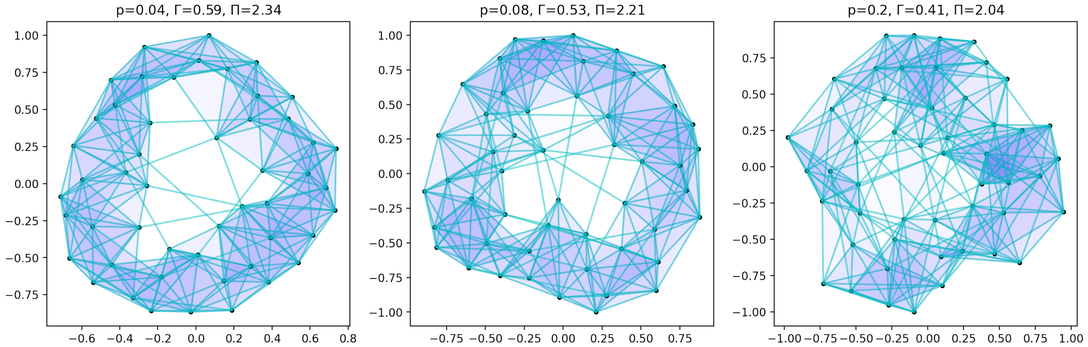
Generalized Laplacian operator

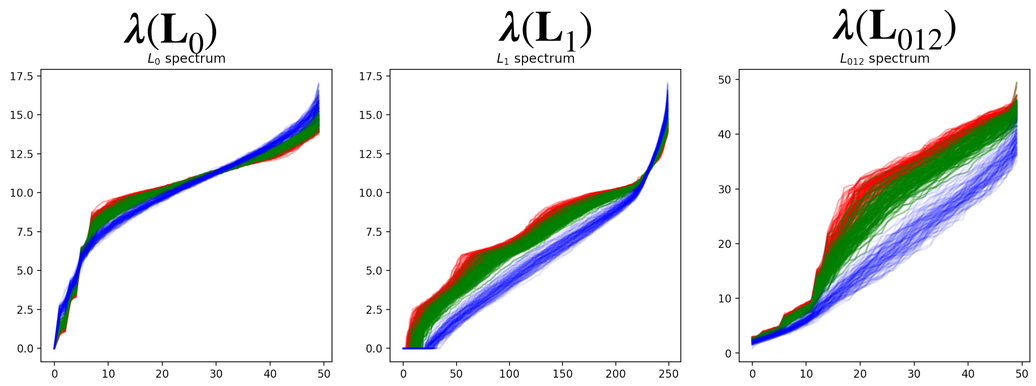
$$\begin{tikzcd} C_3 \arrow[r, "\partial_{3,1}"] \arrow[rr, bend right, "\partial_{3,3}"'] \arrow[rr, bend left, "\partial_{3,2}"] & C_2 \arrow[r, "\partial_{2,1}"] \arrow[r, bend right, "\partial_{2,2}"'] & C_1 \arrow[r, "\partial_{1,1}"] & C_0 \end{tikzcd}$$

$$L_{k,p,q} = \partial_{k,p}^* \partial_{k,p} + \partial_{k+q,q} \partial_{k+q,q}^*$$

Watts-Strogatz model $G(n, m, p)$

$n = 35,\ m = 15,\ p = \{0.01, 0.1, 0.4\}$, 500 graphs of each class



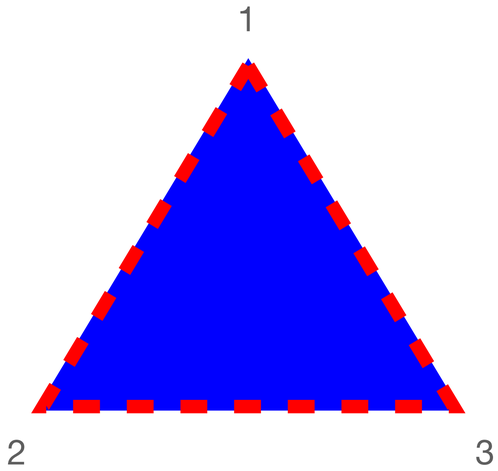


L0	L1	L012
73.91 ± 0.86	78.37 ± 0.62	**84.08 ± 0.49**

Classification accuracy, % for 5-fold cross-validation averaged over 10 runs.

Persistent Laplacian

Triangle



$K \subset L$

$$
\begin{tikzcd}
\Sigma_2^K = 0 & \Sigma_1^K = \{12,13,23\} & \Sigma_0^K = \{1,2,3\} \\
C_{p+1}^K \arrow[r, "\partial_{p+1}^K"] \arrow[d, hook, dashed] & C_p^K \arrow[r, shift right, "\partial_p^K"'] \arrow[d, hook, dashed] \arrow[dl, shift right, "\left(\partial_{p+1}^{L,K}\right)^*"'] & C_{p-1}^K \arrow[l, shift right, "\left(\partial_p^K\right)^*"'] \arrow[d, hook, dashed] \\
C_{p+1}^{L,K} \arrow[ur, shift right, "\partial_p^{L,K}"'] \arrow[dl, hook, dashed] & & \\
C_{p+1}^L \arrow[r, "\partial_{p+1}^L"] & C_p^L \arrow[r, "\partial_p^L"] & C_{p-1}^L \\
\Sigma_2^L = \{123\} & \Sigma_1^L = \{12,13,23\} & \Sigma_0^L = \{1,2,3\}
\end{tikzcd}
$$